Simulation and The Monte Carlo Method

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CHAPTER 1

Systems, Models, Simulation, and the Monte Carlo Methods

In this chapter we discuss the concepts of systems, models, simulation and Monte Carlo methods. This discussion seems necessary in the absence of a unified terminology in the literature. We do not give rigid definitions, however, but explain what we mean when using the above-mentioned terms.

1.1 SYSTEMS

By a system we mean a set of related entities sometimes called *components* or *elements*. For instance, a hospital can be considered as a system, with doctors, nurses, and patients as elements. The elements have certain characteristics, or *attributes*, that have logical or numerical values. In our example an attribute can be, for instance, the number of beds, the number of X-ray machines, skill, quantity, and so on. A number of activities (relations) exist among the elements, and consequently the elements interact. These activities cause changes in the system. For example, the hospital has X-ray machines that have an operator. If there is no operator, the doctors cannot have X-rays of the patients taken.

We consider both internal and external relationships. The internal relationships connect the elements within the system, while the external relationships connect the elements with the environment, that is, with the world outside the system. For instance, an internal relationship is the relationship or interaction between the doctors and nurses, or between

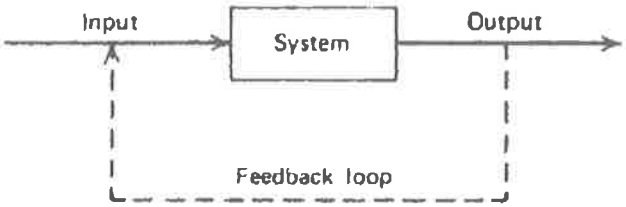

Fig. 1.1.1 Graphical representation of a system.

the nurses and the patients. An external relationship is, for example, the way in which the patients are delivered to the emergency room. We can represent a system by a diagram, as in Fig. 1.1.1.

The system is influenced by the environment through the input it receives from the environment. When a system has the capability of reacting to changes in its own state, we say that the system contains feedback. A nonfeedback, or open-loop, system lacks this characteristic. For an example of feedback consider a waiting line; when there are more than a certain number of patients, the hospital can add more staff to handle the increased workload.

The attributes of the system elements define its *state*. In our example the number of patients waiting for a doctor describe the system's state. When a patient arrives at or leaves the hospital, the system moves to a new state. If the behavior of the elements cannot be predicted exactly, it is useful to take random observations from the probability distributions and to average the performance of the objective. We say that a system is in *equilibrium* or in the *steady state* if the probability of being in some state does not vary in time. There are still actions in the system, that is, the system can still move from one state to another, but the *probabilities* of its moving from one state to another are fixed. These fixed probabilities are limiting probabilities that are realized after a long period of time, and they are independent of the state in which the system started. A system is called *stable* if it returns to the steady state after an external shock in the system. If the system is not in the steady state, it is in a *transient* state.

We can classify systems in a variety of ways. There are *natural* and *artificial systems*, *adaptive* and *nonadaptive systems*. An adaptive system reacts to changes in its environment, whereas a nonadaptive system does not. Analysis of an adaptive system requires a description of how the environment induces a change of state.

Suppose that over a period of time the number of patients increases. If the hospital adds more staff to handle the increased workload, we say that the hospital is an adaptive system.

1.2 MODELS

The first step in studying a system is building a model. The importance of models and model-building has been discussed by Rosenbluth and Wiener [32], who wrote:

> No substantial part of the universe is so simple that it can be grasped and controlled without abstraction. Abstraction consists in replacing the part of the universe under consideration by a model of similar but simpler structure. Models . . . are thus a central necessity of scientific procedure.

A scientific model can be defined as an abstraction of some real system, an abstraction that can be used for prediction and control. The purpose of a scientific model is to enable the analyst to determine how one or more changes in various aspects of the modeled system may affect other aspects of the system or the system as a whole.

A crucial step in building the model is constructing the objective function, which is a mathematical function of the decision variables.

There are many types of models. Churchman et al. [4] and Kiviat [18] described the following kinds:

1 **Iconic models** Those that pictorially or visually represent certain aspects of a system.

2 **Analog models** Those that employ one set of properties to represent some other set of properties that the system being studied possesses.

3 **Symbolic models** Those that require mathematical or logical operations and can be used to formulate a solution to the problem at hand.

In this book, however, we are concerned only with symbolic models (which are also called *abstract* models), that is, we deal with models consisting of mathematical symbols or flowcharts. All other models (iconic, analog, verbal, physical, etc.), although no less important, are excluded from this book.

There are many advantages by using mathematical models. According to Fishman [8] they do the following:

1 Enable investigators to organize their theoretical beliefs and empirical observations about a system and to deduce the logical implications of this organization.

2 Lead to improved system understanding.

3 Bring into perspective the need for detail and relevance.

4 Expedite the analysis.

5 Provide a framework for testing the desirability of system modifications.

6 Allow for easier manipulation than the system itself permits.

7 Permit control over more sources of variation than direct study of a system would allow.

8 Are generally less costly than the system.

An additional advantage is that a mathematical model describes a problem more concisely than, for instance, a verbal description does.

On the other hand, there are at least three reservations in Fishman's monograph [8], which we should always bear in mind while constructing a model.

First, there is no guarantee that the time and effort devoted to modeling will return a useful result and satisfactory benefits. Occasional failures occur because the level of resources is too low. More often, however, failure results when the investigator relys too much on method and not enough on ingenuity; the proper balance between the two leads to the greatest probability of success.

The second reservation concerns the tendency of an investigator to treat his or her particular depiction of a problem as the best representation of reality. This is often the case after much time and effort have been spent and the investigator expects some useful results.

The third reservation concerns the use of the model to predict the range of its applicability without proper qualification.

Mathematical models can be classified in many ways. Some models are *static*, other are *dynamic*. Static models are those that do not explicitly take time-variation into account, whereas dynamic models deal explicitly with time-variable interaction. For instance, Ohm's law is an example of a static model, while Newton's law of motion is an example of a dynamic model.

Another distinction concerns *deterministic* versus *stochastic* models. In a deterministic model all mathematical and logical relationships between the elements are fixed. As a consequence these relationships completely determine the solutions. In a stochastic model at least one variable is random.

While building a model care must be taken to ensure that it remains a valid representation of the problem.

In order to be useful, a scientific model necessarily embodies elements of two conflicting attributes—realism and simplicity. On the one hand, the model should serve as a reasonably close approximation to the real system and incorporate most of the important aspects of the system. On the other hand, the model must not be so complex that it is impossible to understand and manipulate. Being a formalism, a model is necessarily an abstraction.

Often we think that the more details a model includes the better it resembles reality. But adding details makes the solution more difficult and

converts the method for solving a problem from an analytical to an approximate numerical one.

In addition, it is not even necessary for the model to approximate the system to indicate the measure of effectiveness for all various alternatives. All that is required is that there be a high *correlation* between the prediction by the model and what would actually happen with the real system. To ascertain whether this requirement is satisfied or not, it is important to test and establish control over the solution.

Usually, we begin testing the model by re-examining the formulation of the problem and revealing possible flaws. Another criterion for judging the validity of the model is determining whether all mathematical expressions are dimensionally consistent. A third useful test consists of varying input parameters and checking that the output from the model behaves in a plausible manner. The fourth test is the so-called *retrospective* test. It involves using historical data to reconstruct the past and then determining how well the resulting solution would have performed if it had been used. Comparing the effectiveness of this hypothetical performance with what actually happened then indicates how well the model predicts the reality. However, a disadvantage of retrospective testing is that it uses the same data that guided formulation of the model. Unless the past is a true replica of the future, it is better not to resort to this test at all.

Suppose that the conditions under which the model was built change. In this case the model must be modified and control over the solution must be established. Often, it is desirable to identify the critical input parameters of the model, that is, those parameters subject to changes that would affect the solution, and to establish systematic procedures to control them. This can be done by *sensitivity analysis*, in which the respective parameters are varied over their ranges to determine the degree of variation in the solution of the model.

After constructing a mathematical model for the problem under consideration, the next step is to derive a solution from this model. There are *analytic* and *numerical* solution methods.

An analytic solution is usually obtained directly from its mathematical representation in the form of formula.

A numerical solution is generally an approximate solution obtained as a result of substitution of numerical values for the variables and parameters of the model. Many numerical methods are iterative, that is, each successive step in the solution uses the results from the previous step. Newton's method for approximating the root of a nonlinear equation can serve as an example.

Two special types of numerical methods are simulation and the Monte Carlo methods. The following section discusses these.

1.3 SIMULATION AND THE MONTE CARLO METHODS

Simulation has long been an important tool of designers, whether they are simulating a supersonic jet flight, a telephone communication system, a wind tunnel, a large-scale military battle (to evaluate defensive or offensive weapon systems), or a maintenance operation (to determine the optimal size of repair crews).

Although simulation is often viewed as a "method of last resort" to be employed when everything else has failed, recent advances in simulation methodologies, availability of software, and technical developments have made simulation one of the most widely used and accepted tools in system analysis and operations research.

Naylor et al. [28] define simulation as follows:

> Simulation is a numerical technique for conducting experiments on a digital computer, which involves certain types of mathematical and logical models that describe the behavior of business or economic system (or some component thereof) over extended periods of real time.

This definition is extremely broad, however, and can include such seemingly unrelated things as economic models, wind tunnel testing of aircraft, war games, and business management games.

Naylor et al. [28] write:

> The fundamental rationale for using simulation is man's unceasing quest for knowledge about the future. This search for knowledge and the desire to predict the future are as old as the history of mankind. But prior to the seventeenth century the pursuit of predictive power was limited almost entirely to the purely deductive methods of such philosophers as Plato, Aristotle, Euclid, and others.

Simulation deals with both *abstract* and *physical* models. Some simulation with physical and abstract models might involve participation by real people. Examples include link-trainers for pilots and military or business games. Two types of simulation involving real people deserve special mention. One is *operational gaming*, the other *man-machine* simulation.

The term "operational gaming" refers to those simulations characterized by some form of conflict of interest among players or human decision-makers within the framework of the simulated environment, and the experimenter, by observing the players, may be able to test hypotheses concerning the behavior of the individuals and/or the decision system as a whole.

In operational gaming a computer is often used to collect, process, and produce information that human players, usually adversaries, need to make decisions about system operation. Each player's objective is to perform as well as possible. Moreover, each player's decisions affect the information that the computer provides as the game progresses through simulated time. The computer can also play an active role by initiating predetermined or random actions to which the players respond.

War games and business management games are commonly discussed in operational gaming literature (see, e.g., Morgenthaler [23] and Shubik [38]).

Military gaming is essentially a training device for military leaders; it enables them to test the effects of alternative strategies under simulated war conditions. For example, the Naval Electronic Warfare Simulator, developed in the 1950s, consisted of a large analog computer designed primarily to assess ship damage and to provide information to two opposite forces regarding their respective effectiveness in a naval engagement [14, pp. 15, 16]. The exercise, which is one form of simulation gaming, has been used as an educational device for naval fleet officers in the final stages of their training.

Business games are also a type of educational tool, but for training managers or business executives rather than military leaders.

> A business game is a contrived situation which imbeds players in a simulated business environment, where they must make management-type decisions from time to time, and their choices at one time generally affect the environmental conditions under which subsequent decisions must be made. Further, the interaction between decisions and environment is determined by a refereeing process which is not open to argument from the players [30, pp. 7, 8].

In man-machine simulation there is no need for gaming. While interacting with the computer real people in the laboratory perform the data reduction and analysis.

The following two examples are drawn from Fishman [8]:

The Rand Systems Research Laboratory employed simulation to generate stimuli for the study of information processing centers [14, p. 16]. The principal features of a radar site were reproduced in the laboratory, and by carefully controlling the synthetic input to the system and recording the behavior of the human detectors it was possible to examine the relative effectiveness of various man-machine combinations and procedures.

In 1956 Rand established the Logistics System Laboratory under U.S. Air Force sponsorship [10]. The first study in this laboratory involved

simulation of two large logistics systems in order to compare their effectiveness under different management and resource utilization policies. Each system consisted of men and machines, together with policy rules for the use of such resources in simulated stress situations such as war. The simulated environment required a specified number of aircraft in flying and alert states, while the system's capability to meet these objectives was limited by malfunctioning parts, procurement and transportation delays, and the like. The human participants represented management personnel, while higher echelon policies in the utilization of resources were simulated on the computer. The ultimate criteria of the effectiveness of each system were the number of operationally ready aircraft and the dollar cost of maintaining this number.

Although the purpose of the first study in this laboratory was to test the feasibility of introducing new procedures into an existing air force logistics system and to compare the modified system with the original one, the second laboratory problem had quite a different objective. Its purpose was to improve the design of the operational control system through the use of simulation.

Naylor et al. [28] describe many situations where simulation can be successfully used. We mention some of them.

First, it may be either impossible or extremely expensive to obtain data from certain processes in the real world. Such processes might involve, for example, the performance of large-scale rocket engines, the effect of proposed tax cuts on the economy, the effect of an advertising campaign on total sales. In this case we say that the simulated data are necessary to *formulate hypotheses* about the system.

Secondly, the observed system may be so complex that it cannot be described in terms of a set of mathematical equations for which analytic solutions are obtainable. Most economic systems fall into this category. For example, it is virtually impossible to describe the operation of a business firm, an industry, or an economy in terms of a few simple equations. Simulation has been found to be an extremely effective tool for dealing with problems of this type. Another class of problems that leads to similar difficulties is that of large-scale queueing problems involving multiple channels that are either parallel or in series (or both).

Thirdly, even though a mathematical model can be formulated to describe some system of interest, it may not be possible to obtain a solution to the model by straightforward analytic techniques. Again, economic systems and complex queueing problems provide examples of this type of difficulty. Although it may be conceptually possible to use a set of mathematical equations to describe the behavior of a dynamic system

operating under conditions of uncertainty, present-day mathematics and computer technology are simply incapable of handling a problem of this magnitude.

Fourth, it may be either impossible or very costly to perform validating experiments on the mathematical models describing the system. In this case we say that the simulation data can be used to test alternative hypotheses.

In all these cases simulation is the only practical tool for obtaining relevant answers.

Naylor et al. [28] have suggested that simulation analysis might be appropriate for the following reasons:

1 Simulation makes it possible to study and experiment with the complex internal interactions of a given system whether it be a firm, an industry, an economy, or some subsystem of one of these.

2 Through simulation we can study the effects of certain informational, organizational, and environmental changes on the operation of a system by making alterations in the model of the system and observing the effects of these alterations on the system's behavior.

3 Detailed observation of the system being simulated may lead to a better understanding of the system and to suggestions for improving it, suggestions that otherwise would not be apparent.

4 Simulation can be used as a pedagogical device for teaching both students and practitioners basic skills in theoretical analysis, statistical analysis, and decision making. Among the disciplines in which simulation has been used successfully for this purpose are business administration, economics, medicine, and law.

5 Operational gaming has been found to be an excellent means of stimulating interest and understanding on the part of the participant, and is particularly useful in the orientation of persons who are experienced in the subject of the game.

6 The experience of designing a computer simulation model may be more valuable than the actual simulation itself. The knowledge obtained in designing a simulation study frequently suggests changes in the system being simulated. The effects of these changes can then be tested via simulation before implementing them on the actual system.

7 Simulation of complex systems can yield valuable insight into which variables are more important than others in the system and how these variables interact.

8 Simulation can be used to experiment with new situations about which we have little or no information so as to prepare for what may happen.

9 Simulation can serve as a "preservice test" to try out new policies and decision rules for operating a system, before running the risk of experimenting on the real system.

10 Simulations are sometimes valuable in that they afford a convenient way of breaking down a complicated system into subsystems, each of which may then be modeled by an analyst or team that is expert in that area [23, p. 373].

11 Simulation makes it possible to study dynamic systems in either real time, compressed time, or expanded time.

12 When new components are introduced into a system, simulation can be used to help foresee bottlenecks and other problems that may arise in the operation of the system [23, p. 375].

Computer simulation also enables us to *replicate* an experiment. Replication means rerunning an experiment with selected changes in parameters or operating conditions being made by the investigator. In addition, computer simulation often allows us to induce correlation between these random number sequences to improve the statistical analysis of the output of a simulation. In particular a negative correlation is desirable when the results of two replications are to be summed, whereas a positive correlation is preferred when the results are to be differenced, as in the comparison of experiments.

Simulation does not require that a model be presented in a particular format. It permits a considerable degree of freedom so that a model can bear a close correspondence to the system being studied. The results obtained from simulation are much the same as observations or measurements that might have been made on the system itself. To demonstrate the principles involved in executing a discrete simulation, an example of simulating a machine shop is given in Section 1.4. Many programming systems have been developed, incorporating simulation languages. Some of them are general-purpose in nature, while others are designed for specific types of systems. FORTRAN, ALGOL, and PL/1 are examples of general-purpose languages, while GPSS, SIMSCRIPT, and SIMULA are examples of special simulation languages.

Simulation is indeed an invaluable and very versatile tool in those problems where analytic techniques are inadequate. However, it is by no means ideal. Simulation is an imprecise technique. It provides only *statistical estimates* rather than exact results, and it only compares alternatives rather than generating the optimal one. Simulation is also a *slow* and *costly* way to study a problem. It usually requires a large amount of time and great expense for analysis and programming. Finally, simulation yields only *numerical data* about the performance of the system, and sensitivity

analysis of the model parameters is very expensive. The only possibility is to conduct series of simulation runs with different parameter values.

We have defined simulation as a technique of performing *sampling experiments* on the model of the system. This general definition is often called simulation in a *wide sense*, whereas simulation in a *narrow sense*, or *stochastic simulation*, is defined as experimenting with the model over time; it includes sampling stochastic variates from probability distribution [19]. Therefore stochastic simulation is actually a statistical sampling experiment with the model. This sampling involves all the problems of statistical design analysis.

Because sampling from a particular distribution involves the use of random numbers, stochastic simulation is sometimes called *Monte Carlo simulation*. Historically, the Monte Carlo method was considered to be a technique, using random or pseudorandom numbers, for solution of a model. Random numbers are essentially independent random variables uniformly distributed over the unit interval [0, 1]. Actually, what are available at computer centers are arithmetic codes for generating sequences of *pseudorandom digits*, where each digit (0 through 9) occurs with approximately equal probability (likelihood). Consequently, the sequences can model successive flips of a fair ten-side die. Such codes are called *random number generators*. Grouped together, these generated digits yield pseudorandom numbers with any required number of elements. We discuss random and pseudorandom numbers in the next chapter.

One of the earliest problems connected with Monte Carlo method is the famous Buffon's needle problem. The problem is as follows. A needle of length l units is thrown randomly onto a floor composed of parallel planks of equal width d units, where $d > l$. What is the probability that the needle, once it comes to rest, will cross (or touch) a crack separating the planks on the floor? It can be shown that the probability of the needle hitting a crack is $P = 2l/\pi d$, which can be estimated as the ratio of the number of throws hitting the crack to the total number of throws.

In the begining of the century the Monte Carlo method was used to examine the Boltzmann equation. In 1908 the famous statistician Student used the Monte Carlo method for estimating the correlation coefficient in his t-distribution.

The term "Monte Carlo" was introduced by von Neumann and Ulam during World War II, as a code word for the secret work at Los Alamos; it was suggested by the gambling casinos at the city of Monte Carlo in Monaco. The Monte Carlo method was then applied to problems related to the atomic bomb. The work involved direct *simulation* of behavior concerned with random neutron diffusion in fissionable material. Shortly thereafter Monte Carlo methods were used to evaluate complex multidi-

mensional integrals and to solve certain integral equations, occurring in physics, that were not amenable to analytic solution.

The Monte Carlo method can be used not only for solution of stochastic problems, but also for solution of deterministic problems. A deterministic problem can be solved by the Monte Carlo method if it has the same formal expression as some stochastic process. In Chapter 4 we show how the Monte Carlo method can be used for evaluating multidimensional integrals and some parameters of queues and networks. In Chapter 5 the Monte Carlo method is used for solution of certain integral and differential equations.

Another field of application of the Monte Carlo methods is sampling of random variates from probability distributions, which Morgenthaler [23] calls model sampling. Chapter 3 deals with sampling from various distributions.

The Monte Carlo method is now the most powerful and commonly used technique for analyzing complex problems. Applications can be found in many fields from radiation transport to river basin modeling. Recently, the range of applications has been broadening, and the complexity and computational effort required has been increasing, because realism is associated with more complex and extensive problem descriptions.

Finally, we mention some differences between the Monte Carlo method and simulation:

1 In the Monte Carlo method time does not play as substantial a role as it does in stochastic simulation.

2 The observations in the Monte Carlo method, as a rule, are independent. In simulation, however, we experiment with the model over time so, as a rule, the observations are serially correlated.

3 In the Monte Carlo method it is possible to express the response as a rather simple function of the stochastic input variates. In simulation the response is usually a very complicated one and can be expressed explicitly only by the computer program itself.

1.4 A MACHINE SHOP EXAMPLE

This example is quoted from Gordon [11, pp. 570–573]. For better understanding of the example an important distinction to be made is whether an entity is permanent or temporary. Permanent entities can be compactly and efficiently represented in tables, while temporary entities will be volatile records and are usually handled by the list processing technique described later.

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CHAPTER 2

Random Number Generation

2.1 INTRODUCTION

In this chapter we are concerned with methods of generating random numbers on digital computers. The importance of the random numbers in the Monte Carlo method and simulation has been discussed in Chapter 1. The emphasis in this chapter is mainly on the properties of numbers associated with uniform random variates. The term *random number* is used instead of *uniform random number*. Many techniques for generating random numbers have been suggested, tested, and used in recent years. Some of these are based on random phenomena, others on deterministic recurrence procedures.

Initially, *manual methods* were used, including such techniques as coin flipping, dice rolling, card shuffling, and roulette wheels. It was believed that only mechanical (or electronic) devices could yield "truly" random numbers. These methods were too slow for general use, and moreover, sequences generated by them could not be reproduced. Shortly following the advent of the computer it became possible to obtain random numbers with its aid. One method of generating random numbers on a digital computer consists of preparing a table and storing it in the memory of the computer. In 1955 the RAND Corporation published [46] a well known table of a million random digits that may be used in forming such a table. The advantage of this method is reproducibility; its disadvantage is its lack of speed and the risk of exhausting the table.

In view of these difficulties, John von Neumann [56] suggested the *mid-square* method, using the arithmetic operations of a computer. His idea was to take the square of the preceding random number and extract the

middle digits; for example, if we are generating four-digit numbers and arrive at 5232, we square it, obtain 27,373,824; the next number consists of the middle four digits—namely, 3738—and the procedure is repeated. This raises a logical question: how can such sequences, defined in a completely deterministic way, be random? The answer is that they are not really random, but only seem so, and are in fact referred to as *pseudorandom* or *quasi-random*; still we call them *random*, with the appropriate reservation. Von Neumann's method likewise proved slow and awkward for statistical analysis; in addition the sequences tend to cyclicity, and once a zero is encountered the sequence terminates.

We say that the random numbers generated by this or any other method are "good" ones if they are uniformly distributed, statistically independent, and reproducible. A good method is, moreover, necessarily fast and requires minimum memory capacity. Since all these properties are rarely, if ever, realized, some compromise must be found. The congruential methods for generating pseudorandom numbers, discussed in the next section, were designed specifically to satisfy as many of these requirements as possible.

2.2 CONGRUENTIAL GENERATORS

The most commonly used present-day method for generating pseudorandom numbers is one that produces a nonrandom sequence of numbers according to some recursive formula based on calculating the residues modulo of some integer m of a linear transformation. It is readily seen from this definition that each term of the sequence is available in advance, before the sequence is actually generated. Although these processes are completely deterministic, it can be shown [31] that the numbers generated by the sequence appear to be uniformly distributed and statistically independent. Congruential methods are based on a fundamental congruence relationship, which may be expressed as [32]

$$X_{i+1} = (aX_i + c)(\bmod m), \qquad i = 1, \ldots, n, \tag{2.2.1}$$

where the *multiplier* a, the *increment* c, and the *modulus* m are nonnegative integers. The modulo notation $(\bmod m)$ means that

$$X_{i+1} = aX_i + c - mk_i, \tag{2.2.2}$$

where $k_i = [(aX_i + c)/m]$ denotes the largest positive integer in $(aX_i + c)/m$.

Given an initial starting value X_0 (also called the *seed*), (2.2.2) yields a congruence relationship (modulo m) for any value i of the sequence $\{X_i\}$.

Generators that produce random numbers according to (2.2.1) are called *mixed congruential generators*. The random numbers on the unit inverval (0, 1) can be obtained by

$$U_i = \frac{X_i}{m}. \tag{2.2.3}$$

Clearly, such a sequence will repeat itself in at most m steps, and will therefore be periodic. For example, let $a = c = X_0 = 3$ and $m = 5$; then the sequence obtained from the recursive formula $X_{i+1} = 3X_i + 3 (\text{mod } 5)$ is $X_i = 3, 2, 4, 0, 3$.

It follows from (2.2.2) that $X_i < m$ for all i. This inequality means that the period of the generator cannot exceed m, that is, the sequence X_i contains at most m distinct numbers (the period of the generator in the example is 4, while $m = 5$).

Because of the deterministic character of the sequence, the entire sequence recurs as soon as any number is repeated. We say that the sequence "gets into a loop," that is, there is a cycle of numbers that is repeated endlessly. It is shown [31] that all sequences having the form $X_{i+1} = f(X_i)$ "get into a loop." We want, of course, to choose m as large as possible to ensure a sufficiently large sequence of distinct numbers in a cycle.

Let p be the period of the sequence. When p equals its maximum, that is, when $p = m$, we say that the random number generator has a *full period*. It can be shown [31] that the generator defined in (2.2.1) has a full period, m, if and only if:

1. c is *relatively prime* to m, that is, c and m have no common divisor.
2. $a \equiv 1 (\text{mod } g)$ for every prime factor g of m.
3. $a \equiv 1 (\text{mod } 4)$ if m is a multiple of 4.

Condition 1 means that the greatest common divisor of c and m is unity. Condition 2 means that $a = g[a/g] + 1$. Let g be a prime factor of m; then denoting $K = [a/g]$, we may write

$$a = 1 + gk. \tag{2.2.4}$$

Condition 3 means that

$$a = 1 + 4[a/4] \tag{2.2.5}$$

if $m/4$ is an integer.

Greenberger [19] showed that the correlation coefficient between X_i and X_{i+1} lies between the values

$$\frac{1}{a} - \left(\frac{6c}{am}\right)\left(1 - \frac{c}{m}\right) \pm \frac{a}{m},$$

and that its upper bound is achieved when $a = m^{1/2}$ irrespective of the value of c.

Since most computers utilize either a binary or a decimal digit system, we select $m = 2^\beta$ or $m = 10^\beta$, respectively where β denotes the word-length of the particular computer. We discuss both cases separately in the following.

For a binary computer we have from condition 1 that $m = 2^\beta$ guarantees a full period. It follows also from (2.2.1) that, for $m = 2^\beta$, the parameter c must be odd and

$$a = 1 (\text{mod } 4), \tag{2.2.6}$$

which can be achieved by setting

$$a = 2^r + 1, \qquad r \geq 2.$$

It is noted in the literature [25, 35, 44] that good statistical results can be achieved while choosing $m = 2^{35}$, $a = 2^7 + 1$, and $c = 1$.

For a decimal computer $m = 10^\beta$. In order to generate a sequence with a full period, c must be a positive number not divisible by $g = 2$ or $g = 5$, and the multiplier a must satisfy the condition $a \equiv 1 (\text{mod } 20)$, or alternatively, $a = 10^r + 1$, $r > 1$.

Satisfactory statistical results have been achieved [1] by choosing $a = 101$, $c = 1$, $r \geq 4$. In this case X_0 had little or no effect on the statistical properties of the generated sequences.

The second widely used generator is the *multiplicative generator*

$$X_{i+1} = aX_i (\text{mod } m), \tag{2.2.7}$$

which is a particular case of the mixed generator (2.2.1) with $c = 0$.

It can be shown [1, 2, 5, 31] that, generally, a *full* period cannot be achieved here, but a *maximal* period can, provided that X_0 is relatively prime to m and a meets certain congruence conditions.

For a binary computer we again choose $m = 2^\beta$ and it is shown [31] that the maximal period is achieved when $a = 8r \pm 3$. Here r is any positive integer.

The procedure for generating pseudorandom numbers on a binary computer* can be written as:

1 Choose any odd number as a starting value X_0.

2 Choose an integer $a = 8r \pm 3$, where r is any positive integer. Choose a close to $2^{\beta/2}$ (if $\beta = 35$, $a = 2^{17} + 3$ is a good selection).

3 Compute X_1, using fixed point integer arithmetic. This product will consist of 2β bits from which the high-order β bits are discarded, and the low-order β bits represent X_1.

4 Calculate $U_1 = X_1 / 2^\beta$ to obtain a uniformly distributed variable.

*This procedure and the one that follows are reproduced almost verbatim from Ref. 31.